

DECODE Scoring and Strategy Background

What Do Teams Need to Know Before Choosing a Strategy?

In DECODE, two alliances compete at the same time. Each alliance has two robots. Robots try to score game pieces, build patterns, avoid penalties, and return to their base before time runs out.

This course uses a simplified version of the full FTC challenge. Your team will still use the real ideas of engineering strategy: choosing what to attempt, designing under constraints, testing with evidence, and explaining why your choices make sense.

The Big Idea

A strong robot is not always the robot that tries to do the most things. A strong robot is one that can do useful things reliably.

For this course, teams should think about four strategy questions:

- How will we score balls or artifacts?
- Where will we launch from?
- What can we do in autonomous?
- How will we help both alliance robots park or return to base?

Course Constraints

These constraints shape your strategy.

Constraint	What It Means
No intake	Your robot will not collect balls from the field. A human operator will load balls into the robot.
Hopper limit goal	Your hopper should be designed to carry up to three balls.
No AprilTag reading	Your robot will not use camera vision to identify field targets.
No distance sensors	Your robot will not automatically measure exact distance from the goal.
Limited build time	A simple reliable robot may outperform an unfinished complex robot.
Shared field space	Your robot should leave room for an alliance partner to move, score, and park.

These are not failures. They are engineering constraints. Good teams make smart choices inside real limits.

Match Periods

FTC matches have two main periods.

Period	What Happens
Autonomous	Robots run pre-programmed instructions without driver control.
TeleOp	Human drivers control the robots.

In the official DECODE game, autonomous can include moving off the launch line, scoring artifacts, and building a pattern. In this course, every team should try for a simple autonomous movement. Some teams may challenge themselves to attempt autonomous scoring if their robot is ready.

Scoring Ideas That Matter for This Course

The official DECODE game includes several ways to earn points. Your team does not need to master every official rule to make a strong strategy, but you do need to understand the parts that affect your design.

Scoring Idea	Why It Matters
Scoring artifacts or balls	Your launcher, hopper, driver control, and launch site all affect scoring.
Leaving the launch line in autonomous	A simple autonomous movement can create reliable early points.
Returning to base	Parking or returning to base can be a valuable endgame goal.
Two robots fully returned to base	Alliances can earn an additional bonus when both robots fully fit in base.
Ranking points	Some achievements can matter beyond match points because they help teams rank higher.
Penalties	A risky strategy can lose value if it creates penalties.

Why Parking and Robot Size Matter

In DECODE, both robots may need to return to the alliance base at the end of the match. If both robots can fully fit, the alliance can earn an additional bonus.

This makes robot size a strategy decision. A team might choose:

- a narrow robot that leaves room for an alliance partner,
- a compact robot that can park quickly,
- a taller or unusual robot shape that tries to share space differently,
- or a wider robot that is easier to build but may make parking harder.

The unusual choice is not automatically better. A narrow or tall robot may create problems with balance, hopper space, launcher placement, battery access, and driver control. The team should only choose that path if the parking benefit is worth the trade-off.

Human Operator Loading

Because this course does not include an intake, your human operator matters.

Your team should decide:

- where the human operator loads balls,
- how the robot lines up for loading,
- how many balls the hopper can safely hold,
- how the driver knows the robot is ready to leave,

- and how the team avoids unsafe or illegal loading.

A fast launcher does not help if the robot takes too long to load or if the hopper jams.

Launch Site Choice

Because your robot will not use AprilTag reading or distance sensors, your team needs a repeatable launch site.

A good launch site should be:

- legal,
- reachable,
- easy for the driver to recognize,
- far enough from traffic or defense,
- compatible with the launcher angle and power,
- and respectful of the alliance partner's path.

Your team should choose one first launch site and one backup launch site.

Autonomous Goals

Every team should try to create at least one autonomous program that moves the robot predictably.

Possible autonomous goals:

Goal	Difficulty	Why a Team Might Choose It
Drive off the launch line or starting area	Easier	Reliable first autonomous success.
Drive to a useful position	Medium	Sets up TeleOp or avoids traffic.
Park or move toward base	Medium	Supports a simple endgame or field-position plan.
Score autonomously	Hard	Strong challenge if the launcher and starting setup are consistent.

Autonomous scoring is exciting, but it is risky without sensors. Small changes in starting position, battery level, wheel traction, or ball placement can change the result.

Penalty Avoidance

Avoiding penalties is part of strategy. A team can lose the value of a good scoring plan by breaking rules, blocking illegally, loading unsafely, contacting where contact is not allowed, or continuing an action after time expires.

Teams should ask:

- What are the easiest penalties for us to accidentally commit?
- What should the driver avoid?
- What should the human operator avoid?
- What should the coach remind the team about?
- What part of our strategy needs a rules check before we rely on it?

Strategy Trade-Offs

Good strategy means choosing what not to do.

If We Prioritize...	It Helps Us...	But It May Make It Harder To...
Reliable three-ball scoring	Build and practice one clear plan	Score more than one cycle
Multiple scoring cycles	Increase scoring potential	Stay accurate, avoid traffic, and avoid penalties
A fixed launch site	Aim more consistently	Recover if the site is blocked
Narrow parking-friendly robot	Fit with an alliance partner	Fit the hopper, launcher, battery, and electronics
Simple autonomous movement	Earn reliable early value	Attempt autonomous scoring
Autonomous scoring	Create strong challenge evidence	Finish other systems on time
Penalty avoidance	Protect the score	Drive aggressively or attempt risky plays

Evidence Teams Should Collect

A good strategy should produce evidence. During testing, collect information such as:

- number of balls scored out of three,
- time needed to load, drive, aim, and shoot,
- number of complete scoring cycles in a match-length test,
- accuracy from the first launch site and backup launch site,
- whether two robots can fully fit in base,
- autonomous success rate across several trials,
- driver feedback about control and aiming,
- and rule or penalty concerns noticed during practice.

Main Takeaway

Your team does not need the most complicated robot. Your team needs a strategy that fits your materials, time, drivers, human operator, alliance partner, and evidence from testing.

The best engineering explanation sounds like this:

We chose this strategy because it gave us the best chance to score reliably, avoid penalties, and support our alliance. We accepted this trade-off, and our testing evidence showed this was the right next step.